Engg. Mathematics-1 (BAS-103)

GATEWAY CLASSES

Unit-1 : Syllabus

As per New Syllabus 2022-23

Matrices

- Elementary transformations
- Inverse of a matrix
- Rank of matrix
- Solution of system of linear equations
- Characteristic equation, Cayley-Hamilton Theorem and its application
- Linear Dependence and Independence of vectors
- Eigen values and Eigen vectors
- Complex Matrices, Hermitian, Skew-Hermitian and Unitary Matrices,
- Applications to Engineering problems.

Engineering Mathematics-I

BAS-103

Lec-1

As per
New Syllabus
2022-23

Today's Target

- ✓ Matrix
- ✓ Types of Matrices

By Kumar Gulshan Sir

Matrices

- A Matrix is a rectangular array (or arrangement in rows or columns) of numbers or functions.
- Such an arrangement is enclosed by [ ] or ().

3 X 3

For example A = $\begin{bmatrix} 2 & 3 & 4 \\ 4 & 3 & 1 \\ 1 & 2 & 4 \end{bmatrix}$ R_1 R_2 R_3

C_1 C_2 C_3

Order of Matrix :

- Order = Row x Column

Matrices

An m x n matrix is usually written as

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \dots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \dots & a_{2n} \\ a_{31} & a_{32} & a_{33} & \dots & a_{3n} \\ \dots & \dots & \dots & \dots & \dots \\ a_{m1} & a_{m2} & a_{m3} & \dots & a_{mn} \end{bmatrix} = [a_{ij}]_{m \times n}$$

$m \times n$

Types of Matrices

1. Square Matrix . A matrix in which the number of rows = number of column is called square matrix.

Example. A = $\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$

Trace = $a_{11} + a_{22} + a_{33}$

2. Diagonal Matrix . A square matrix in which all non-diagonal elements are zero is called a diagonal matrix.

Example

$$A = \begin{bmatrix} 2 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$A = \begin{bmatrix} 2 & -1 & 0 \end{bmatrix}$$

3. Scalar Matrix . A diagonal matrix in which **all diagonal elements are equal** to a scalar, say k, is called scalar matrix.

Example. $A = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$

4. Unit Matrix or Identity Matrix . A scalar matrix in which each diagonal element is unity (i.e. , 1) is called unit matrix or Identity Matrix .

Example. $I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$$AI = A$$

$$I^n = I$$

5. Null Matrix . A matrix in which **each element is zero** is called null matrix or void matrix.

Example. $A = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

$AO = O$

6. Transpose of a Matrix . If in a given matrix A, we interchange the rows and the corresponding columns, the **new matrix** obtained is called the transpose of the matrix A.

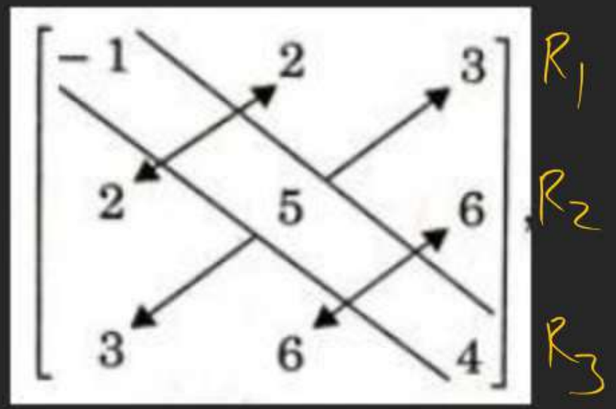
Example. $A = \begin{bmatrix} 2 & 3 & 4 \\ 4 & 3 & 1 \\ 1 & 2 & 4 \end{bmatrix} \begin{matrix} R_1 \\ R_2 \\ \end{matrix}$

$A' = \begin{bmatrix} 2 & 4 & 1 \\ 3 & 3 & 2 \\ 4 & 1 & 4 \end{bmatrix}$

$C_1 \quad C_2 \quad C_3$

7. Symmetric Matrix . A square matrix is called symmetric matrix if

$$A' = A$$

Example. A = 

$$A' = \begin{bmatrix} -1 & 2 & 3 \\ 2 & 5 & 6 \\ 3 & 6 & 4 \end{bmatrix}$$

8. Skew Symmetric Matrix . A square matrix is called skew symmetric matrix if $A' = -A$

Example. $A = \begin{bmatrix} 0 & 2 & -3 \\ -2 & 0 & 1 \\ 3 & -1 & 0 \end{bmatrix} \begin{matrix} R_1 \\ R_2 \\ \\ \end{matrix}$

$$A' = \begin{bmatrix} 0 & -2 & 3 \\ 2 & 0 & -1 \\ -3 & 1 & 0 \end{bmatrix}$$

$$= -\begin{bmatrix} 0 & 2 & -3 \\ -2 & 0 & 1 \\ 3 & -1 & 0 \end{bmatrix}$$

$$\boxed{A' = -A}$$

9. Orthogonal Matrix . A square matrix is called an orthogonal matrix

if $AA' = A'A = I$ or $A' = A^{-1}$

10. Nilpotent Matrix . A square matrix is called a nilpotent matrix if

if $A^2 = 0$

$A^2 = A \times A$

11. Involutary Matrix . A square matrix is called an **Involutary** matrix

if $A^2 = I$

12. Triangular Matrix . A square matrix in which all the elements either **below** or **above** the principal diagonal is zero is called **triangular matrix**.

$$\begin{bmatrix} 2 & 3 & 4 \\ 0 & -1 & 5 \\ 0 & 0 & 3 \end{bmatrix}$$

Upper triangular

$$\begin{bmatrix} -1 & 0 & 0 \\ 5 & 6 & 0 \\ 3 & 2 & 0 \end{bmatrix}$$

Lower triangular

13. Row Matrix. Any matrix having only **ONE row** and any number of columns is called row matrix.

Example. $[2 \quad 5 \quad -3 \quad 0]$

14. Column Matrix. Any matrix having only **ONE column** and any number of rows is called column matrix.

Example. $\begin{bmatrix} \sqrt{2} \\ 0 \\ -1 \end{bmatrix}$

15.Equal Matrices . Two matrices **A** and **B** are said to be equal if

(i) They are of same order

(ii) The elements in the corresponding position are equal.

$$A = \begin{bmatrix} 2 & 3 \\ 4 & 1 \end{bmatrix}_{2\times 2}$$

$$B = \begin{bmatrix} 2 & 3 \\ 4 & 1 \end{bmatrix}_{2\times 2}$$

16.Singular Matrices . A matrix 'A' is said to be singular if

$$|A| = 0.$$

$$A = \begin{bmatrix} 2 & -3 & 4 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{bmatrix} \text{ is a singular matrix since } |A| = 0.$$

17.Non-singular Matrices . A matrix 'A' is said to non – singular matrix if

$$|A| \neq 0.$$

Differences between matrix and determinant

Matrix

- A matrix has no numerical value.

It is only a convenient way of storing information.

- The no. of rows and column may or may not be equal.

Determinant

- A determinant has a numerical value.

- The no. of rows = no. of column

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Engineering Mathematics-I

BAS-103

Lec-2

As per
New Syllabus
2022-23

Today's Target

Operations on Matrices

- ✓ Addition and Subtraction
- ✓ Scalar Multiplication
- ✓ Multiplication of matrices
- ✓ Determinants

By Kumar Gulshan Sir

Addition and Subtraction

Two matrices can be added or subtracted if they have the same order.

For example if $A = \begin{bmatrix} 2 & 5 & -1 \\ 3 & 0 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & -6 & 2 \\ -2 & 5 & 7 \end{bmatrix}$ **the A + B ? and A – B =?**

2X3 2X3

$$A+B = \begin{bmatrix} 2 & 5 & -1 \\ 3 & 0 & 4 \end{bmatrix} + \begin{bmatrix} 1 & -6 & 2 \\ -2 & 5 & 7 \end{bmatrix}$$

$$= \begin{bmatrix} 2+1 & 5-6 & -1+2 \\ 3-2 & 0+5 & 4+7 \end{bmatrix}$$

$$= \begin{bmatrix} 3 & -1 & 1 \\ 1 & 5 & 11 \end{bmatrix}$$

$$A-B = \begin{bmatrix} 2 & 5 & -1 \\ 3 & 0 & 4 \end{bmatrix} - \begin{bmatrix} 1 & -6 & 2 \\ -2 & 5 & 7 \end{bmatrix}$$

$$= \begin{bmatrix} 2-1 & 5-(-6) & -1-2 \\ 3-(-2) & 0-5 & 4-7 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 11 & -3 \\ 5 & -5 & -3 \end{bmatrix}$$

Multiplication of a matrix by a scalar

$$A = \begin{bmatrix} 2 & 0 & 6 \\ 1 & 5 & 2 \\ 5 & 7 & 1 \end{bmatrix}_{3 \times 3}$$

$$4A = 4 \begin{bmatrix} 2 & 0 & 6 \\ 1 & 5 & 2 \\ 5 & 7 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 8 & 0 & 24 \\ 4 & 20 & 8 \\ 20 & 28 & 4 \end{bmatrix}$$

Multiplication of matrices

AB

The product of two matrices is possible only if

Number of columns in first matrix = Number of rows in second matrix

① $A_{m\times n} \times B_{n\times p} = (AB)_{m\times p}$

② $A_{3\times 2} \times B_{3\times 2}$ X

③ $A_{3\times 2} \times B_{2\times 3} = (AB)_{3\times 3}$

④ $A_{4\times 3} \times B_{3\times 2} = (AB)_{4\times 2}$

Question : If $A = \begin{bmatrix} 1 & -2 & 3 \\ 2 & 3 & -1 \\ -3 & 1 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 2 \\ 1 & 2 & 0 \end{bmatrix}$ then determine AB.

Sol. $AB = \begin{bmatrix} 1 & -2 & 3 \\ 2 & 3 & -1 \\ -3 & 1 & 2 \end{bmatrix} \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 2 \\ 1 & 2 & 0 \end{bmatrix}$

R_1C_1 R_1C_2 R_1C_3

R_2C_1 R_2C_2 R_2C_3

R_3C_1 R_3C_2 R_3C_3

$$AB = \begin{bmatrix} 1+0+3 & 0-2+6 & 2-4+0 \\ 2+0-1 & 0+3-2 & 4+6+0 \\ -3+0+2 & 0+1+4 & -6+2+0 \end{bmatrix}$$

$$AB = \begin{bmatrix} 4 & 4 & -2 \\ 1 & 1 & 10 \\ -1 & 5 & -4 \end{bmatrix}$$

Determinant

Question : Find the determinant of the following matrix $A = \begin{bmatrix} 2 & 7 \\ 3 & 10 \end{bmatrix}_{2\times 2}$

$$|A| = \begin{vmatrix} 2 & 7 \\ 3 & 10 \end{vmatrix}$$

$$= 2 \times 10 - 3 \times 7$$

$$= 20 - 21$$

$$= -1$$

Determinant

Question : Find the determinant of the following matrix $A = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 3 & 1 & 1 \end{bmatrix}$.

3 X 3

$$|A| = \begin{vmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 3 & 1 & 1 \end{vmatrix}$$

$$\begin{vmatrix} + & - & + \\ - & + & - \\ + & - & + \end{vmatrix}$$

Expand along R_1

$$|A| = 0\begin{vmatrix} 2 & 3 \\ 1 & 1 \end{vmatrix} - 1\begin{vmatrix} 1 & 3 \\ 3 & 1 \end{vmatrix} + 2\begin{vmatrix} 1 & 2 \\ 3 & 1 \end{vmatrix}$$

$$= 0 - 1(1-9) + 2(1-6)$$

$$= 0 - 1(-8) + 2(-5)$$

$$0 + 8 - 10$$

$$= -2$$

$$\boxed{|A| = -2}$$

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Engineering Mathematics-1

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Unit-1

Matrices

Today's Target

Elementary Transformation

Lec-3

- ✓ Inverse Elementary Transformation
- ✓ AKTU Questions
- ✓ Practice Questions

Elementary Transformation

Elementary Row Transformation

(i) Interchanging of two rows

$$A = \begin{bmatrix} 3 & 1 & 4 \\ 2 & -3 & 4 \\ 1 & -1 & 1 \end{bmatrix} \begin{matrix} R_1 \\ R_2 \\ R_3 \end{matrix}$$

$R_1 \longleftrightarrow R_3$

$$A \sim \begin{bmatrix} 1 & -1 & 1 \\ 2 & -3 & 4 \\ 3 & 1 & 4 \end{bmatrix}$$

Elementary Column Transformation

(i) Interchanging of two columns

$$A = \begin{bmatrix} 3 & 1 & 4 \\ 2 & -3 & 4 \\ 1 & -1 & 1 \end{bmatrix}$$

$C_1 \quad C_2 \quad C_3$

$C_1 \longleftrightarrow C_2$

$$A \sim \begin{bmatrix} 1 & 3 & 4 \\ -3 & 2 & 4 \\ -1 & 1 & 1 \end{bmatrix}$$

Elementary Transformation

(ii) Multiplication of a row by a non-zero number

$$A = \begin{bmatrix} 3 & 1 & 4 \\ 2 & -3 & 4 \\ 1 & -1 & 1 \end{bmatrix} \begin{matrix} R_1 \\ R_2 \\ R_3 \end{matrix}$$

$R_2 \longrightarrow 2R_2$

$$A \sim \begin{bmatrix} 3 & 1 & 4 \\ 4 & -6 & 8 \\ 1 & -1 & 1 \end{bmatrix}$$

(ii) Multiplication of column by a non-zero number

$$A = \begin{bmatrix} 3 & 1 & 4 \\ 2 & -3 & 4 \\ 1 & -1 & 1 \end{bmatrix}$$

$C_1 \quad C_2 \quad C_3$

$C_2 \longrightarrow 4C_2$

$$A \sim \begin{bmatrix} 3 & 4 & 4 \\ 2 & -12 & 4 \\ 1 & -4 & 1 \end{bmatrix}$$

Elementary Transformation

(iii) Addition of k times the element of a row to corresponding elements of another row, $k \neq 0$

$$A = \begin{bmatrix} 3 & 1 & 4 \\ 2 & -3 & 4 \\ 1 & -1 & 1 \end{bmatrix} \begin{matrix} R_1 \\ R_2 \\ R_3 \end{matrix}$$

$$R_2 \longrightarrow R_2 + (-2R_3)$$

$$A \sim \begin{bmatrix} 3 & 1 & 4 \\ 0 & -1 & 2 \\ 1 & -1 & 1 \end{bmatrix}$$

(iii) Addition of k times the element of column to corresponding elements of another column, $k \neq 0$

$$A = \begin{bmatrix} 3 & 1 & 4 \\ 2 & -3 & 4 \\ 1 & -1 & 1 \end{bmatrix}$$

$C_1 \quad C_2 \quad C_3$

3	2	1
-3	9	3
0	11	4

$$C_1 \longrightarrow C_1 + (-3C_2)$$

$$A \sim \begin{bmatrix} 0 & 1 & 4 \\ 11 & -3 & 4 \\ 4 & -1 & 1 \end{bmatrix}$$

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Ques.1 Transform $\begin{bmatrix} 1 & 3 & 3 \\ 2 & 4 & 10 \\ 3 & 8 & 4 \end{bmatrix}$ into a unit matrix by using elementary transformation.

3 X 3

$$\begin{bmatrix} 1 & 3 & 3 \\ 2 & 4 & 10 \\ 3 & 8 & 4 \end{bmatrix}$$

$$R_2 \longrightarrow R_2 + (-2R_1)$$

$$R_3 \longrightarrow R_3 + (-3R_1)$$

$$\sim \begin{bmatrix} 1 & 3 & 3 \\ 0 & -2 & 4 \\ 0 & -1 & -5 \end{bmatrix}$$

$$R_2 \longrightarrow -\frac{1}{2} R_2$$

$$\sim \begin{bmatrix} 1 & 3 & 3 \\ 0 & 1 & -2 \\ 0 & -1 & -5 \end{bmatrix}$$

$$R_1 \longrightarrow R_1 + (-3R_2)$$

$$R_3 \longrightarrow R_3 + R_2$$

$$\sim \begin{bmatrix} 1 & 0 & 9 \\ 0 & 1 & -2 \\ 0 & 0 & -7 \end{bmatrix}$$

$$R_3 \longrightarrow -\frac{1}{7} R_3$$

$$\sim \begin{bmatrix} 1 & 0 & 9 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{bmatrix}$$

$$R_1 \longrightarrow R_1 + (-9R_3)$$

$$R_2 \longrightarrow R_2 + 2R_3$$

$$\sim \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Question.2 Find inverse employing elementary transformation $A = \begin{bmatrix} 3 & -3 & 4 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{bmatrix}$

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$A = IA$

$$\begin{bmatrix} 3 & -3 & 4 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} A$$

$R_1 \longrightarrow R_1 - R_2$

$$\begin{bmatrix} 1 & 0 & 0 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} A$$

$R_2 \longrightarrow R_2 + (-2R_1)$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & -3 & 4 \\ 0 & -1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -1 & 0 \\ -2 & 3 & 0 \\ 0 & 0 & 1 \end{bmatrix} A$$

$R_2 \longleftrightarrow R_3$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 1 \\ 0 & -3 & 4 \end{bmatrix} = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 0 & 1 \\ -2 & 3 & 0 \end{bmatrix} A$$

$R_2 \longrightarrow -R_2$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & -3 & 4 \end{bmatrix} = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 0 & -1 \\ -2 & 3 & 0 \end{bmatrix}$$

$R_3 \longrightarrow R_3 + 3R_2$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 0 & -1 \\ -2 & 3 & -3 \end{bmatrix} A$$

$R_2 \longrightarrow R_2 + R_3$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -1 & 0 \\ -2 & 3 & -4 \\ -2 & 3 & -3 \end{bmatrix} A$$

$$\therefore A^{-1} = \begin{bmatrix} 1 & -1 & 0 \\ -2 & 3 & -4 \\ -2 & 3 & -3 \end{bmatrix}$$

Ques.3 Compute the inverse of the matrix $\begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 3 & 1 & 1 \end{bmatrix}$ by employing elementary row transformation .

AKTU

$$A = IA$$

$$\begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 3 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} A$$

$R_1 \longleftrightarrow R_2$

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 3 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} A$$

$R_3 \longrightarrow R_3 + (-3R_1)$

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & -5 & -8 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & -3 & 1 \end{bmatrix} A$$

$R_1 \longrightarrow R_1 + (-2R_2)$

$R_3 \longrightarrow R_3 + 5R_2$

$$\begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \\ 0 & 0 & 2 \end{bmatrix} = \begin{bmatrix} -2 & 1 & 0 \\ 1 & 0 & 0 \\ 5 & -3 & 1 \end{bmatrix} A$$

$R_3 \longrightarrow \frac{1}{2}R_3$

$$\begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} -2 & 1 & 0 \\ 1 & 0 & 0 \\ \frac{5}{2} & -\frac{3}{2} & \frac{1}{2} \end{bmatrix} A$$

$R_1 \longrightarrow R_1 + R_3$

$R_2 \longrightarrow R_2 + (-2R_3)$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ -4 & 3 & -1 \\ \frac{5}{2} & -\frac{3}{2} & \frac{1}{2} \end{bmatrix} A$$

$$A^{-1} = \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ -4 & 3 & -1 \\ \frac{5}{2} & -\frac{3}{2} & \frac{1}{2} \end{bmatrix}$$

Practice Q .1 Find the inverse of the matrix $A = \begin{bmatrix} 8 & 4 & 3 \\ 2 & 1 & 1 \\ 1 & 2 & 1 \end{bmatrix}$

Practice Q .2 Find the inverse of the matrix $A = \begin{bmatrix} 1 & 2 & 1 \\ 3 & 2 & 3 \\ 1 & 1 & 3 \end{bmatrix}$

Practice Q .3 Find the inverse of the matrix $A = \begin{bmatrix} 2 & 3 & 4 \\ 4 & 3 & 1 \\ 1 & 2 & 4 \end{bmatrix}$

Thank You

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Engineering Mathematics-1

Lec-4

Unit-1
Matrices

Today's Target

Rank of Matrix

- ✓ Rank ???
- ✓ Echelon Form method
- ✓ AKTU Questions
- ✓ Practice Questions

Rank of a Matrix

Rank = Number of non-zero rows in a matrix

Note:

(i) If A is a null matrix, then $\rho(A) = 0$

$A = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}_{2\times 2}$

(ii) If A is NOT a null matrix, then $\rho(A) \geq 1$

(iii) If A is non singular n x n matrix then $|A| \neq 0$

$\rho(A) = n$

(iv) If A is m x n matrix then

$\rho(A) \leq$ minimum of m and n

① $A_{2\times 3}$

$\rho(A) \leq 2$

② $A_{3\times 2}$

$\rho(A) \leq 2$

③ $A_{3\times 4}$

$\rho(A) \leq 3$

Question 4.1 Find the value of 'b' so that rank of $A = \begin{bmatrix} 2 & 4 & 2 \\ 3 & 1 & 2 \\ 1 & 0 & b \end{bmatrix}$ is 2.

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$$A = \begin{bmatrix} 2 & 4 & 2 \\ 3 & 1 & 2 \\ 1 & 0 & b \end{bmatrix}$$

$$|A| = \begin{vmatrix} 2 & 4 & 2 \\ 3 & 1 & 2 \\ 1 & 0 & b \end{vmatrix} \quad (+ \; - \; +)$$

$$|A| = 2\begin{vmatrix} 1 & 2 \\ 0 & b \end{vmatrix} - 4\begin{vmatrix} 3 & 2 \\ 1 & b \end{vmatrix} + 2\begin{vmatrix} 3 & 1 \\ 1 & 0 \end{vmatrix}$$

$$|A| = 2(b-0) - 4(3b-2) + 2(0-1)$$

$$|A| = 2b - 12b + 8 - 2$$

$$|A| = -10b + 6$$

$$-10b + 6 = 0$$

$$+10b = +6$$

$$b = \frac{6}{+10} \; {}^{3}_{5}$$

$$\boxed{b = \frac{3}{5}}$$

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Question 4.2 Find the value of 'P' so that rank of A = $\begin{bmatrix} 3 & P & P \\ P & 3 & P \\ P & P & 3 \end{bmatrix}$ is 1.

$$A = \begin{bmatrix} 3 & P & P \\ P & 3 & P \\ P & P & 3 \end{bmatrix}$$

$$|A| = \begin{vmatrix} 3 & P \\ P & 3 \end{vmatrix}$$

$$\begin{vmatrix} 3 & P \\ P & 3 \end{vmatrix} = 0$$

$$9 - P^2 = 0$$

$$+P^2 = +9$$

$$P^2 = 9$$

$$\boxed{P = 3}$$

Practice Q. 4.1 Find the value of 'P' so that rank of A = $\begin{bmatrix} 1 & 2 & 3 \\ 3 & -1 & 2 \\ 4 & 1 & K \end{bmatrix}$ is 2.

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Ans. K = 5

Echelon Form method

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Rank = Number of non-zero rows in upper triangular matrix

Question 4.3 Find the rank of the matrix $A = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 4 & 2 \\ 2 & 6 & 5 \end{bmatrix}$.

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 4 & 2 \\ 2 & 6 & 5 \end{bmatrix} \begin{matrix} R_1 \\ R_2 \\ R_3 \end{matrix}$$

$R_2 \longrightarrow R_2 + (-R_1)$

$R_3 \longrightarrow R_3 + (-2R_1)$

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 2 & -1 \\ 0 & 2 & -1 \end{bmatrix}$$

$R_3 \longrightarrow R_3 + (-R_2)$

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 2 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

Non zero Row ①

Non-zero row ②

Zero Row

$\rho(A) = 2$

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Question 4.4 Find the rank of the matrix $A = \begin{bmatrix} 2 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{bmatrix}$.

$$A = \begin{bmatrix} 2 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{bmatrix} \begin{matrix} R_1 \\ R_2 \\ R_3 \end{matrix} \quad 3 \times 3$$

$$R_2 \longrightarrow R_2 + (-R_1)$$

$$R_3 \longrightarrow R_3 + (-R_1)$$

$$A = \begin{bmatrix} 2 & 2 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{matrix} \text{Non-zero row } (1) \\ \text{zero row} \\ \text{zero row} \end{matrix}$$

$$\boxed{\rho(A) = 1}$$

AKTU

Practice Q. 4.2 Find the rank of the matrix $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 7 \\ 3 & 6 & 10 \end{bmatrix}$.

Ans. 2

AKTU

Question 4.5 Find the rank of the matrix $A = \begin{bmatrix} 1 & 2 & 3 & 2 \\ 2 & 3 & 5 & 1 \\ 1 & 3 & 4 & 5 \end{bmatrix}$.

3×4

$$A = \begin{bmatrix} 1 & 2 & 3 & 2 \\ 2 & 3 & 5 & 1 \\ 1 & 3 & 4 & 5 \end{bmatrix}$$

$R_2 \longrightarrow R_2 + (-2R_1)$ $\quad R_3 \longrightarrow R_3 + (-R_1)$

$$A = \begin{bmatrix} 1 & 2 & 3 & 2 \\ 0 & -1 & -1 & -3 \\ 0 & 1 & 1 & 3 \end{bmatrix}$$

$R_3 \longrightarrow R_3 + R_2$

$$A = \begin{bmatrix} 1 & 2 & 3 & 2 \\ 0 & -1 & -1 & -3 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{matrix} R_1 \\ R_2 \\ R_3 \end{matrix}$$

$\rho(A) = 2$

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Practice Q. 4.3 Find the rank of the matrix $A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \\ 4 & 3 & 1 & 2 \end{bmatrix}$.

Ans. 3

Question 4.6 Use elementary transformation to reduce the following matrix A to triangular form and hence find the rank of $A = \begin{bmatrix} 2 & 3 & -1 & -1 \\ 1 & -1 & -2 & -4 \\ 3 & 1 & 3 & -2 \\ 6 & 3 & 0 & -7 \end{bmatrix}$.

$$A = \begin{bmatrix} 2 & 3 & -1 & -1 \\ 1 & -1 & -2 & -4 \\ 3 & 1 & 3 & -2 \\ 6 & 3 & 0 & -7 \end{bmatrix} \begin{matrix} R_1 \\ R_2 \\ R_3 \\ R_3 \end{matrix}$$

$R_1 \longleftrightarrow R_2$

$$A \sim \begin{bmatrix} 1 & -1 & -2 & -4 \\ 2 & 3 & -1 & -1 \\ 3 & 1 & 3 & -2 \\ 6 & 3 & 0 & -7 \end{bmatrix}$$

$R_2 \longrightarrow R_2 + (-2R_1)$

$R_3 \longrightarrow R_3 + (-3R_1)$

$R_4 \longrightarrow R_4 + (-6R_1)$

$$A \sim \begin{bmatrix} 1 & -1 & -2 & -4 \\ 0 & 5 & 3 & 7 \\ 0 & 4 & 9 & 10 \\ 0 & 9 & 12 & 17 \end{bmatrix}$$

$R_2 \longrightarrow R_2 + (-R_3)$

$$A \sim \begin{bmatrix} 1 & -1 & -2 & -4 \\ 0 & 1 & -6 & -3 \\ 0 & 4 & 9 & 10 \\ 0 & 9 & 12 & 17 \end{bmatrix}$$

$R_3 \longrightarrow R_3 + (-4R_2)$

$R_4 \longrightarrow R_4 + (-9R_2)$

$$A \sim \begin{bmatrix} 1 & -1 & -2 & -4 \\ 0 & 1 & -6 & -3 \\ 0 & 0 & 33 & 22 \\ 0 & 0 & 66 & 44 \end{bmatrix}$$

$R_4 \longrightarrow R_4 + (-2R_3)$

$$A \sim \begin{bmatrix} 1 & -1 & -2 & -4 \\ 0 & 1 & -6 & -3 \\ 0 & 0 & 33 & 22 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{matrix} ① \\ ② \\ ③ \\ X \end{matrix}$$

$\rho(A) = 3$

Practice Q. 4.4 Use elementary transformation to reduce the following matrix A to triangular form and hence find the rank of A = $\begin{bmatrix} -1 & 2 & 3 & -2 \\ 2 & -5 & 1 & 2 \\ 3 & -8 & 5 & 2 \\ 5 & -12 & -1 & 6 \end{bmatrix}$.

Ans. 2

Thank You

Engineering Mathematics-1

GATEWAY CLASSES

Lec-6

Unit-1
Matrices

Today's Target

Rank of Matrix

- ✓ Normal form method (Part-2)
- ✓ AKTU Questions
- ✓ Practice Questions

Question 6.1 Reduce the matrix A to its normal form and hence find its rank where

$$A = \begin{bmatrix} 1 & -1 & 2 & -3 \\ 4 & 1 & 0 & 2 \\ 0 & 3 & 0 & 4 \\ 0 & 1 & 0 & 2 \end{bmatrix}_{4\times 4}$$

$$A = \begin{bmatrix} 1 & -1 & 2 & -3 \\ 4 & 1 & 0 & 2 \\ 0 & 3 & 0 & 4 \\ 0 & 1 & 0 & 2 \end{bmatrix}$$

$R_2 \longrightarrow R_2 + (-4R_1)$

$$\sim \begin{bmatrix} 1 & -1 & 2 & -3 \\ 0 & 5 & -8 & 14 \\ 0 & 3 & 0 & 4 \\ 0 & 1 & 0 & 2 \end{bmatrix}$$

$C_2 \longrightarrow C_2 + C_1$

$C_3 \longrightarrow C_3 + (2C_4)$

$C_4 \longrightarrow C_4 + 3C_4$

$$\sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 5 & -8 & 14 \\ 0 & 3 & 0 & 4 \\ 0 & 1 & 0 & 2 \end{bmatrix} \begin{matrix} R_1 \\ R_2 \\ R_3 \\ R_4 \end{matrix}$$

$R_2 \longleftrightarrow R_4$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 2 \\ 0 & 3 & 0 & 4 \\ 0 & 5 & -8 & 14 \end{bmatrix}$$

$R_3 \longrightarrow R_3 + (-3R_2)$

$R_4 \longrightarrow R_4 + (-5R_2)$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 0 & -2 \\ 0 & 0 & -8 & 4 \end{bmatrix}$$

$C_4 \longrightarrow C_4 + (-2C_2)$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & -2 \\ 0 & 0 & -8 & 4 \end{bmatrix}$$

$C_3 \longleftrightarrow C_4$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -2 & 0 \\ 0 & 0 & 4 & -8 \end{bmatrix}$$

$R_3 \longrightarrow -\frac{1}{2} R_3$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 4 & -8 \end{bmatrix}$$

$R_4 \longrightarrow R_4 + (-4R_3)$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -8 \end{bmatrix}$$

$C_4 \longrightarrow -\frac{1}{8} C_4$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$\rho(A) = 4$

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Ques 6.2 If $A = \begin{bmatrix} 3 & -3 & 4 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{bmatrix}_{3\times3}$, Determine two non-singular matrices P and Q such that

PAQ = I. Hence find A^{-1}.

$A = I_3 A I_3$

$$\begin{bmatrix} 3 & -3 & 4 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} A \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$R_1 \longrightarrow R_1 - R_2$

$$\begin{bmatrix} 1 & 0 & 0 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} A \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$R_2 \longrightarrow R_2 + (-2R_1)$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & -3 & 4 \\ 0 & -1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -1 & 0 \\ -2 & 3 & 0 \\ 0 & 0 & 1 \end{bmatrix} A \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$R_2 \longleftrightarrow R_3$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 1 \\ 0 & -3 & 4 \end{bmatrix} = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 0 & 1 \\ -2 & 3 & 0 \end{bmatrix} A \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$R_2 \longrightarrow -R_2$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & -3 & 4 \end{bmatrix} = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 0 & -1 \\ -2 & 3 & 0 \end{bmatrix} A \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$R_3 \longrightarrow R_3 + 3R_2$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 0 & -1 \\ -2 & 3 & -3 \end{bmatrix} A \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$C_3 \longrightarrow C_3 + C_2$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & -1 & 0 \\ 0 & 0 & -1 \\ -2 & 3 & -3 \end{bmatrix}}_{P} A \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}}_{Q}$$

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$$P = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 0 & -1 \\ -2 & 3 & -3 \end{bmatrix}$$

$$Q = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

$$I = PAQ$$

$$P^{-1}I = (P^{-1}P)AQ$$

$$IP^{-1} = IAQ$$

$$P^{-1} = AQ$$

$$(P^{-1}P) = AQP$$

$$I = AQP$$

$$A^{-1}I = (A^{-1}A)QP$$

$$\boxed{A^{-1} = QP}$$

$$A^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 & 0 \\ 0 & 0 & -1 \\ -2 & 3 & -3 \end{bmatrix}$$

$$= \begin{bmatrix} 1+0+0 & -1+0+0 & 0+0+0 \\ 0+0-2 & 0+0+3 & 0-1-3 \\ 0+0-2 & 0+0+3 & 0+0-3 \end{bmatrix}$$

$$\therefore A^{-1} = \begin{bmatrix} 1 & -1 & 0 \\ -2 & 3 & -4 \\ -2 & 3 & -3 \end{bmatrix}$$

$$\therefore \ldots$$

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Practice Q. 6.1 For the matrix A $\begin{bmatrix} 1 & 1 & 2 \\ 1 & 2 & 3 \\ 0 & -1 & -1 \end{bmatrix}$ Find non-singular matrices P and Q such that PAQ is a normal form.

Ans. $P = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ -1 & 1 & 1 \end{bmatrix}$ $Q = \begin{bmatrix} 1 & -1 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}$

Practice Q 6.2 Find non-singular matrices P and Q such that PAQ is a normal form for the matrix $A = \begin{bmatrix} 1 & -1 & -1 \\ 1 & 1 & 1 \\ 3 & 1 & 1 \end{bmatrix}$.

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Ans. $P = \begin{bmatrix} 1 & 0 & 0 \\ -\frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{2} & 1 & -\frac{1}{2} \end{bmatrix}$ $Q = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}$

Thank You

Engineering Mathematics-1

Lec-7

Unit-1
Matrices

Today's Target

Solution of System of Linear Equations

- ✓Non-Homogenous Linear Equations
- ✓Homogenous Linear Equations
- ✓AKTU Questions
- ✓Practice Questions

$$[A:B]=\begin{bmatrix} a_1 & b_1 & c_1 : d_1 \\ a_2 & b_2 & c_2 : d_2 \\ a_3 & b_3 & c_3 : d_2 \end{bmatrix}$$

System of Linear Equations

Non-Homogenous System of Linear Equations

$$a_1x + b_1y + c_1z = d_1$$
$$a_2x + b_2y + c_2z = d_2$$
$$a_3x + b_3y + c_3z = d_3$$

$$X=\begin{bmatrix} x \\ y \\ z \end{bmatrix} \quad B=\begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix}$$

$$A=\begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix}$$

$$AX = B$$

Homogenous System of Linear Equations

$$A=\begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix}$$

$$a_1x + b_1y + c_1z = 0$$
$$a_2x + b_2y + c_2z = 0$$
$$a_3x + b_3y + c_3z = 0$$

$$X=\begin{bmatrix} x \\ y \\ z \end{bmatrix} \quad B=\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$AX = 0$$

A System of non-Homogenous Linear Equations

$AX = B$

Find $\rho(A)$ and $\rho[A : B]$

$\rho(A) = \rho[A : B]$

Solution exists, system is consistent

$\rho(A) \neq \rho[A : B]$

No Solution, system is inconsistent

$\rho(A) = \rho[A : B]$
$= n$ (no. of unknowns)

Unique Solution

$\rho(A) = \rho[A : B] < n$ (no. of unknowns)

Infinite no. of Solution

Q.7.1 Solve, with the help of matrices, the simultaneous equations:

$$x + y + z = 3, \quad x + 2y + 3z = 4, \quad x + 4y + 9z = 6$$

$x + y + z = 3$

$x + 2y + 3z = 4$

$x + 4y + 9z = 6$

$$[A : B] = \begin{bmatrix} 1 & 1 & 1 & : & 3 \\ 1 & 2 & 3 & : & 4 \\ 1 & 4 & 9 & : & 6 \end{bmatrix}$$

$R_2 \longrightarrow R_2 - R_1$

$R_3 \longrightarrow R_3 - R_1$

$$[A : B] = \begin{bmatrix} 1 & 1 & 1 & : & 3 \\ 0 & 1 & 2 & : & 1 \\ 0 & 3 & 8 & : & 3 \end{bmatrix}$$

$R_3 \longrightarrow R_3 + (-3R_2)$

$$[A : B] = \begin{bmatrix} 1 & 1 & 1 & : & 3 \\ 0 & 1 & 2 & : & 1 \\ 0 & 0 & 2 & : & 0 \end{bmatrix}$$

$\rho(A) = 3 \qquad \rho[A : B] = 3$

$\therefore \rho(A) = \rho[A : B] =$ No. of unknowns

The given system of equations is consistent with Unique solution

$x + y + z = 3 \quad -(1)$

$y + 2z = 1 \quad -(2)$

$2z = 0 \quad -(3)$

$z = 0$

$y = 1$

$x + 1 + 0 = 3$

$x = 2$

Q.7.2 Investigate for consistency of the following equations and if possible find the solution.

$$4x - 2y + 6z = 8$$
$$x + y - 3z = -1$$
$$15x - 3y + 9z = 21$$

$4x - 2y + 6z = 8$

$x + y - 3z = -1$

$15x - 3y + 9z = 21$

$$[A : B] = \begin{bmatrix} 4 & -2 & 6 & : & 8 \\ 1 & 1 & -3 & : & -1 \\ 15 & -3 & 9 & : & 21 \end{bmatrix}$$

$R_1 \longleftrightarrow R_2$

$$\sim \begin{bmatrix} 1 & 1 & -3 & : & -1 \\ 4 & -2 & 6 & : & 8 \\ 15 & -3 & 9 & : & 21 \end{bmatrix}$$

$R_2 \longrightarrow R_2 + (-4R_1)$

$R_3 \longrightarrow R_3 + (-15R_1)$

$$\sim \begin{bmatrix} 1 & 1 & -3 & : & -1 \\ 0 & -6 & 18 & : & 12 \\ 0 & -18 & 54 & : & 36 \end{bmatrix}$$

$R_3 \longrightarrow R_3 + (-3R_2)$

$$\sim \begin{bmatrix} 1 & 1 & -3 & : & -1 \\ 0 & -6 & 18 & : & 12 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$$

$\rho(A) = 2 \quad \rho[A : B] = 2$

$\rho(A) = \rho[A : B] = 2$ ($<$ No. of variables)

The system is consistant with infinite solution

$x + y - 3z = -1$ — (1)

$-6y + 18z = 12$ — (2)

Let $z = k$

$-6y + 18k = 12$

$-6y = -18k + 12$

$y = 3k - 3$

$x + y - 3z = -1$

$x + 3k - 2 - 3k = -1$

$x - 2 = -1$

$x = 2 - 1$

$x = 1$

Practice Q 7.1. Test for consistency and solve the following system of equations:

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$$2x - y + 3z = 8$$

$$-x + 2y + z = 4$$

$$3x + y - 4z = 0$$

Ans. X = 2, y = 2, z = 2

Q7.3 Show that the system of equations $x + y + z = -3$, $3x + y - 2z = -2$, $2x + 4y + 7z = 7$ is not consistent.

$$x + y + z = -3$$
$$3x + y - 2z = -2$$
$$2x + 4y + 7z = 7$$

$$(A:B)\begin{bmatrix} 1 & 1 & 1 & : & -3 \\ 3 & 1 & -2 & : & -2 \\ 2 & 4 & 7 & : & 7 \end{bmatrix}$$

$R_2 \longrightarrow R_2 + (-3R_1)$

$R_3 \longrightarrow R_3 + (-2R_1)$

$$\sim \begin{bmatrix} 1 & 1 & 1 & : & -3 \\ 0 & -2 & -5 & : & 7 \\ 0 & 2 & 5 & : & 3 \end{bmatrix}$$

$R_3 \longrightarrow R_3 + R_2$

$$\sim \begin{bmatrix} 1 & 1 & 1 & : & -3 \\ 0 & -2 & -5 & : & 7 \\ 0 & 0 & 0 & : & 10 \end{bmatrix}$$

$\rho(A) = 2$

$\rho(A:B) = 3$

$\therefore \rho(A) \neq \rho(A:B)$

The given system of equation is not consistent (Inconsistent)

No solution

Q7.4 Determine, for what values of λ and μ the following system of equations

$$x+y+z=6,\quad x+2y+3z=10,\quad x+2y+\lambda z=\mu$$

AKTU-2016,18

have (i) no solution (ii) unique solution (iii) infinite solutions

$x+y+z=6$

$x+2y+3z=10$

$x+2y+\lambda z=\mu$

$$[A:B]=\begin{bmatrix}1 & 1 & 1 & : & 6\\ 1 & 2 & 3 & : & 10\\ 1 & 2 & \lambda & : & \mu\end{bmatrix}$$

$R_2 \to R_2-R_1$

$R_3 \to R_3-R_1$

$$\sim\begin{bmatrix}1 & 1 & 1 & : & 6\\ 0 & 1 & 2 & : & 4\\ 0 & 1 & \lambda-1 & : & \mu-6\end{bmatrix}$$

$R_3 \to R_3-R_2$

$$\sim\begin{bmatrix}1 & 1 & 1 & : & 6\\ 0 & 1 & 2 & : & 4\\ 0 & 0 & \lambda-3 & : & \mu-10\end{bmatrix}$$

(i) For no solution

$\lambda-3=0$

$\lambda=3$

$\mu-10\neq 0$

$\mu\neq 10$

(ii) For unique solution

$\lambda-3\neq 0$

$\lambda\neq 3$

μ can have any value

(iii) For infinite solution

$\lambda-3=0$ | $\mu-10=0$

$\lambda=3$ | $\mu=10$

Practice Q 7.2. Determine the values of λ and μ such that the equations

$$2x - 5y + 2z = 8, \quad 2x + 4y + 6z = 5, \quad x + 2y + \lambda z = \mu$$

have (i) no solution (ii) a unique solution (iii) infinite solutions

Ans. (i) $\lambda = 3$ and $\mu \neq 5/2$ (ii) $\lambda \neq 3$ and μ may have any value (iii) $\lambda = 3$ and $\mu = 5/2$

A System of Homogenous Linear Equations

$AX = 0$

Always has a Solution

Find $\rho(A)$

$\rho(A) = n$ (no. of unknowns)

Unique or trivial Solution (each unknown equal to zero)

$\rho(A) < n$ (no. of unknowns)

Infinite no. of non-trivial Solutions

Q7.5 Determine the values of λ so that the equations $2x + y + 2z = 0$, $x + y + 3z = 0$, $4x + 3y + \lambda z = 0$ have a non-zero solution.

$$2x + y + 2z = 0$$
$$x + y + 3z = 0$$
$$4x + 3y + \lambda z = 0$$

$$A = \begin{bmatrix} 2 & 1 & 2 \\ 1 & 1 & 3 \\ 4 & 3 & \lambda \end{bmatrix}$$

$R_1 \leftrightarrow R_2$

$$A \sim \begin{bmatrix} 1 & 1 & 3 \\ 2 & 1 & 2 \\ 4 & 3 & \lambda \end{bmatrix}$$

$R_2 \to R_2 + (-2R_1)$

$R_3 \to R_3 + (-4R_1)$

$$A \sim \begin{bmatrix} 1 & 1 & 3 \\ 0 & -1 & -4 \\ 0 & -1 & \lambda - 12 \end{bmatrix}$$

$R_3 \to R_3 - R_2$

$$\sim \begin{bmatrix} 1 & 1 & 3 \\ 0 & -1 & -4 \\ 0 & 0 & \lambda - 8 \end{bmatrix}$$

For non-zero solution

$$\lambda - 8 = 0$$

$$\boxed{\lambda = 8}$$

OR

$$|A| = \begin{vmatrix} 2 & 1 & 2 \\ 1 & 1 & 3 \\ 4 & 3 & \lambda \end{vmatrix}$$

$$\lambda - 2 - 6 + 12 = 0$$
$$\lambda - 8 = 0$$
$$\boxed{\lambda = 8}$$

For non-zero solution

$$|A| = 0$$

$$\begin{vmatrix} 2 & 1 & 2 \\ 1 & 1 & 3 \\ 4 & 3 & \lambda \end{vmatrix} = 0$$

Expand along first row

$$2\begin{vmatrix} 1 & 3 \\ 3 & \lambda \end{vmatrix} - 1\begin{vmatrix} 1 & 3 \\ 4 & \lambda \end{vmatrix} + 2\begin{vmatrix} 1 & 1 \\ 4 & 3 \end{vmatrix} = 0$$

$$2(\lambda - 9) - 1(\lambda - 12) + 2(3 - 4) = 0$$

$$2\lambda - 18 - \lambda + 12 - 2 = 0$$

Practice Q 7.3. Using matrix method, show that the equations

$$3x + 3y + 2z = 1, \quad x + 2y = 4, \quad 10y + 3z = -2, \quad 2x - 3y = 5$$

are consistent and hence obtain the solution for x, y and z.

Ans. X = 2, y = 1, z = -4

Thank You

Engineering Mathematics-1

Lec-8

Unit-1

Matrices

Today's Target

Linear Dependence and Linear Independence of Vectors

✓AKTU Questions

✓Practice Questions

Linear Dependence and Linear Independence of Vectors

Vectors(Matrices) $X_1, X_2 \ldots\ldots X_n$ are said to be dependent if

$X_1 = (2 \quad 1 \quad 3)$

- ✓ **All the vectors are of same order**
- ✓ **n scalars $\lambda_1, \lambda_2 \ldots\ldots \lambda_n$ exists such that**

$$\lambda_1 X_1 + \lambda_2 X_2 + \cdots + \lambda X_n = 0$$

otherwise vectors are Independent

Linear Dependence and Linear Independence of Vectors

Procedure to test linear dependence of vectors

✓ **Step 1.** Construct coefficient matrix A with elements of given vectors as columns.

✓ **Step 2.** Find ρ(A).

✓ **Step 3.** If ρ(A) = no. of vectors then given set of vectors is linearly independent.

If ρ(A) < no. of vectors then given set of vectors is linearly dependent.

Q. 8.1 Show that the vectors $X_1 = (1, -1, 1)$, $X_2 = (2, 1, 1)$, $X_3 = (3, 0, 2)$ are linearly dependent. Find the relation between them.

$X_1 = (1, -1, 1)$

$X_2 = (2, 1, 1)$

$X_3 = (3, 0, 2)$

$$A = \begin{bmatrix} 1 & 2 & 3 \\ -1 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix}$$

$R_2 \longrightarrow R_2 + R_1$

$R_3 \longrightarrow R_3 + (-R_1)$

$$\sim \begin{bmatrix} 1 & 2 & 3 \\ 0 & 3 & 3 \\ 0 & -1 & -1 \end{bmatrix}$$

$R_2 \longrightarrow \frac{1}{3} R_2$

$$\sim \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 1 \\ 0 & -1 & -1 \end{bmatrix}$$

$R_3 \longrightarrow R_3 + R_2$

$$\sim \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} \begin{matrix} R_1 \\ R_2 \end{matrix}$$

$\rho(A) = 2$ (< No. of vectors)

$\therefore X_1, X_2, X_3$ are linearly dependent

$a_1 + 2a_2 + 3a_3 = 0$

$a_2 + a_3 = 0$

$a_2 = -a_3$

$\frac{a_2}{1} = \frac{a_3}{-1} = K$

$a_2 = K$

$a_3 = -K$

$a_1 + 2K - 3K = 0$

$a_1 - K = 0$

$a_1 = K$

$a_1 X_1 + a_2 X_2 + a_3 X_3 = 0$

$KX_1 + KX_2 - KX_3 = 0$

$KX_1 + KX_2 = KX_3$

$X_1 + X_2 = X_3$

Practice Q.8.1 Examine the following vectors for linear dependence and find the relation between them, if possible:

$X_1 = (1, 1, -1, 1)$, $X_2 = (1, -1, -2, -1)$, $X_3 = (3, 1, 0, 1)$

Ans. Linearly dependent; $2X_1 + X_2 = X_3$

Q. 8.2 Find the value of λ for which the vectors (1, -2, λ), (2, -1, 5), and (3, -5, 7λ) are linearly dependent.

$X_1 = (1, -2, \lambda)$

$X_2 = (2, -1, 5)$

$X_3 = (3, -5, 7\lambda)$

$$A = \begin{bmatrix} 1 & 2 & 3 \\ -2 & -1 & -5 \\ \lambda & 5 & 7\lambda \end{bmatrix}$$

∵ Vectors are dependent

$\rho(A) < 3$

It is possible when

$|A| = 0$

$$\begin{vmatrix} 1 & 2 & 3 \\ -2 & -1 & -5 \\ \lambda & 5 & 7\lambda \end{vmatrix} = 0$$

$1(-7\lambda + 25) - 2(-14\lambda + 5\lambda) + 3(-10 + \lambda) = 0$

$-7\lambda + 25 - 2(-9\lambda) - 30 + 3\lambda = 0$

$-7\lambda + 25 + 18\lambda - 30 + 3\lambda = 0$

$14\lambda - 5 = 0$

$$\boxed{\lambda = \frac{5}{14}}$$

Practice Q.8.2 Show that the vectors $X_1 = (1, 2, 4)$, $X_2 = (2, -1, 3)$, $X_3 = (0, 1, 2)$ and $X_4 = (-3, 7, 2)$ are linearly dependent and find the relation between them.

Ans. Linearly dependent; $9X_1 - 12X_2 + 5X_3 - 5X_4 = 0$

Q. 8.3 Find whether or not the following vectors are linearly dependent or independent (1, 1, 1, 1), (0, 1, 1, 1), (0, 0, 1, 1), (0, 0, 0, 1).

$X_1 = (1, 1, 1, 1)$

$X_2 = (0, 1, 1, 1)$

$X_3 = (0, 0, 1, 1)$

$X_4 = (0, 0, 0, 1)$

$$A = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 \end{bmatrix}$$

$R_2 \to R_2 - R_1$

$R_3 \to R_3 - R_1$

$R_4 \to R_4 - R_1$

$$\sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 \end{bmatrix}$$

$R_3 \to R_3 - R_2$

$R_4 \to R_4 - R_2$

$$\sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

$R_4 \to R_4 - R_3$

$$\sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$\rho(A) = 4 (=$ No. of vectors$)$

$\therefore$ The given vectors are linearly dependent

Practice Q.8.3 Find whether or not the following vectors are linearly dependent or independent (1, 1, 0), (1, 0, 1), (0, 1, 1).

Ans. $\rho(A) = 2$ (<3, no of vectors)

Hence the given set of vectors is linearly dependent.

Thank You

Engineering Mathematics-1

Lec-9

Unit-1

Matrices

Today's Target

Complex Matrices

- ✓ AKTU Questions
- ✓ Practice Questions

Complex Matrices

If at least one element of a matrix is a complex number, then the matrix is called complex matrix.

$$A = \begin{bmatrix} 1 & -1 & 0 \\ 1 & 1 & i \\ 3 & 1 & 2 \end{bmatrix} \qquad A = \begin{bmatrix} 2+i & 3 & -1+3i \\ -5 & i & 4-2i \end{bmatrix} \qquad P = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ -1 & 1 & 1 \end{bmatrix}$$

Complex Number : $z = x + iy$ Where x and y are real numbers and

$i = \sqrt{-1}$

$i^2 = -1$

Real Imaginary

$\bar{z} = x - iy$

Conjugate of Matrix

$$A = \begin{bmatrix} 1 & -i & 1+i \\ i & 0 & 2-3i \\ 1-i & 2+3i & 2 \end{bmatrix}$$

$$\overline{A} = \begin{bmatrix} 1 & i & 1-i \\ -i & 0 & 2+3i \\ 1+i & 2-3i & 2 \end{bmatrix}$$

$$\overline{\overline{A}} = A$$

$$\overline{AB} = \overline{A}\,\overline{B}$$

$$\overline{A+B} = \overline{A} + \overline{B}$$

$$\overline{\lambda A} = \overline{\lambda}\,\overline{A}$$

Transpose of Conjugate of Matrix

Or

Conjugate of Transpose of Matrix

$$(\overline{A})' = (\overline{A'}) = A^{\theta}$$

$$A = \begin{bmatrix} 2 & 1-i \\ 3+i & 3 \end{bmatrix}$$

$$\overline{A} = \begin{bmatrix} 2 & 1+i \\ 3-i & 3 \end{bmatrix}$$

$$(\overline{A})' = \begin{bmatrix} 2 & 3-i \\ 1+i & 3 \end{bmatrix}$$

$$A' = \begin{bmatrix} 2 & 3+i \\ 1-i & 3 \end{bmatrix}$$

$$\overline{(A')} = \begin{bmatrix} 2 & 3-i \\ 1+i & 3 \end{bmatrix}$$

Hermitian Matrix

✓ A Square matrix A is said to be Hermitian if $A = A^{\theta}$

$$A = \begin{bmatrix} 2 & 2-i \\ 2+i & 3 \end{bmatrix}$$

Skew-Hermitian Matrix

✓ A Square matrix A is said to be skew-Hermitian if $A^{\theta} = -A$

$$A = \begin{bmatrix} 0 & 2+i \\ -2+i & 0 \end{bmatrix}$$

Unitary Matrix

✓ A Square matrix A is said to be unitary if $A^{\theta}A = AA^{\theta} = I$

① $(A^{\theta})^{\theta} = A$

② $(AB)^{\theta} = B^{\theta}A^{\theta}$

③ $(A+B)^{\theta} = A^{\theta} + B^{\theta}$

④ $(\lambda A)^{\theta} = \bar{\lambda} A^{\theta}$

Note: $A^{-1} = A^{\theta}$

Q. 9.1 If A is a skew-Hermitian matrix, then show that iA is Hermitian.

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$$A^{\theta} = -A$$

$$(iA)^{\theta} = \bar{i}\, A^{\theta}$$

$$(iA)^{\theta} = (-i)(-A)$$

$$(iA)^{\theta} = iA$$

$\therefore$ iA is a Hermitian matrix

Q. 9.2 Show that the matrix A is Hermitian and iA is skew- Hermitian where A is

$$A = \begin{bmatrix} 2 & 3-4i \\ 3+4i & 2 \end{bmatrix}$$

$$\overline{A} = \begin{bmatrix} 2 & 3+4i \\ 3-4i & 2 \end{bmatrix}$$

$$A^{\theta} = (\overline{A})' = \begin{bmatrix} 2 & 3-4i \\ 3+4i & 2 \end{bmatrix}$$

$$\boxed{A^{\theta} = A}$$

$\therefore$ A is a Hermitian matrix

$$iA = i\begin{bmatrix} 2 & 3-4i \\ 3+4i & 2 \end{bmatrix}$$

$$= \begin{bmatrix} 2i & 3i-4i^2 \\ 3i+4i^2 & 2i \end{bmatrix}$$

$$iA = \begin{bmatrix} 2i & 3i+4 \\ 3i-4 & 2i \end{bmatrix}$$

$$\overline{iA} = \begin{bmatrix} -2i & 4-3i \\ -4-3i & -2i \end{bmatrix}$$

$$(\overline{iA})' = \begin{bmatrix} -2i & -4-3i \\ 4-3i & -2i \end{bmatrix}$$

$$(iA)^{\theta} = \begin{bmatrix} -2i & -4-3i \\ 4-3i & -2i \end{bmatrix}$$

$$= -\begin{bmatrix} 2i & 4+3i \\ -4+3i & 2i \end{bmatrix}$$

$$\boxed{(iA)^{\theta} = -iA}$$

$\therefore$ iA is a skew Hermitian matrix

Practice Q.9.1 Show that the matrix A is Hermitian and iA is skew-

Hermitian where A is $\begin{bmatrix} 2 & 3+2i & -4 \\ 3-2i & 5 & 6i \\ -4 & -6i & 3 \end{bmatrix}$.

Q. 9.3 Define unitary matrix. Show that the following matrix is unitary:

$$A = \frac{1}{\sqrt{3}}\begin{bmatrix} 1 & 1+i \\ 1-i & -1 \end{bmatrix}.$$

$$\bar{A} = \frac{1}{\sqrt{3}}\begin{bmatrix} 1 & 1-i \\ 1+i & -1 \end{bmatrix}$$

$$A^\theta = (\bar{A})' = \frac{1}{\sqrt{3}}\begin{bmatrix} 1 & 1+i \\ 1-i & -1 \end{bmatrix}$$

$$AA^\theta = \frac{1}{\sqrt{3}}\begin{bmatrix} 1 & 1+i \\ 1-i & -1 \end{bmatrix}\frac{1}{\sqrt{3}}\begin{bmatrix} 1 & 1+i \\ 1-i & -1 \end{bmatrix}$$

$$= \frac{1}{3}\begin{bmatrix} 1+1^2-i^2 & 1+i-1(1+i) \\ 1-i-1(1-i) & i^2-i^2+1 \end{bmatrix}$$

$$= \frac{1}{3}\begin{bmatrix} 1+1+1 & 1+i-1-i \\ 1-i-1+i & 1+1+1 \end{bmatrix}$$

$$AA^\theta = \frac{1}{3}\begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix}$$

$$= \frac{3}{3}\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$AA^\theta = I$$

Practice Q.9.2 If $A = \begin{bmatrix} 2+i & 3 & -1+3i \\ -5 & i & 4-2i \end{bmatrix}$, verify that A*A is a Hermitian matrix where A* is the conjugate transpose of A.

Q. 9.4 Show that the matrix $\begin{bmatrix} \alpha + i\gamma & -\beta + i\delta \\ \beta + i\delta & \alpha - i\gamma \end{bmatrix}$ is unitary if $\alpha^2 + \beta^2 + \gamma^2 + \delta^2 = 1$

$$A = \begin{bmatrix} \alpha + i\gamma & -\beta + i\delta \\ \beta + i\delta & \alpha - i\gamma \end{bmatrix}$$

$$\bar{A} = \begin{bmatrix} \alpha - i\gamma & -\beta - i\delta \\ \beta - i\delta & \alpha + i\gamma \end{bmatrix}$$

$$A^\theta = \begin{bmatrix} \alpha - i\gamma & \beta - i\delta \\ -\beta - i\delta & \alpha + i\gamma \end{bmatrix}$$

$$AA^\theta = \begin{bmatrix} \alpha + i\gamma & -\beta + i\delta \\ \beta + i\delta & \alpha - i\gamma \end{bmatrix} \begin{bmatrix} \alpha - i\gamma & \beta - i\delta \\ -\beta - i\delta & \alpha + i\gamma \end{bmatrix}$$

$$= \begin{bmatrix} \alpha^2 - \alpha\gamma i + \alpha\delta i - \gamma^2 i^2 + \beta^2 + \beta\gamma i - \beta\delta i - i^2\delta^2 & 0 \\ 0 & \beta^2 - \beta\gamma i + \beta\gamma i - i^2\delta^2 \\ & \alpha^2 + \alpha\gamma i - \alpha\gamma i - i^2\gamma^2 \end{bmatrix}$$

$$= \begin{bmatrix} \alpha^2 + \gamma^2 + \beta^2 + \delta^2 & 0 \\ 0 & \beta^2 + \delta^2 + \alpha^2 + \gamma^2 \end{bmatrix}$$

$$AA^\theta = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\boxed{AA^\theta = I}$$

Q. 9.5 Show that $A = \frac{1}{\sqrt{3}}\begin{bmatrix} 1 & 1 & 1 \\ 1 & \omega & \omega^2 \\ 1 & \omega^2 & \omega \end{bmatrix}$ is a unitary matrix, where ω is complex cube root of unity.

$$A = \frac{1}{\sqrt{3}}\begin{bmatrix} 1 & 1 & 1 \\ 1 & \omega & \omega^2 \\ 1 & \omega^2 & \omega \end{bmatrix}$$

$$\bar{A} = \frac{1}{\sqrt{3}}\begin{bmatrix} 1 & 1 & 1 \\ 1 & \omega^2 & \omega \\ 1 & \omega & \omega^2 \end{bmatrix}$$

$$A^\theta = \frac{1}{\sqrt{3}}\begin{bmatrix} 1 & 1 & 1 \\ 1 & \omega^2 & \omega \\ 1 & \omega & \omega^2 \end{bmatrix}$$

$$A A^\theta = \frac{1}{\sqrt{3}}\begin{bmatrix} 1 & 1 & 1 \\ 1 & \omega & \omega^2 \\ 1 & \omega^2 & \omega \end{bmatrix}\frac{1}{\sqrt{3}}\begin{bmatrix} 1 & 1 & 1 \\ 1 & \omega^2 & \omega \\ 1 & \omega & \omega^2 \end{bmatrix}$$

$$= \frac{1}{3}\begin{bmatrix} 1+1+1 & 1+\omega^2+\omega & 1+\omega+\omega^2 \\ 1+\omega+\omega^2 & 1+\omega^3+\omega^3 & 1+\omega^2+\omega^4 \\ 1+\omega^2+\omega & 1+\omega^4+\omega^2 & 1+\omega^3+\omega^3 \end{bmatrix}$$

$$= \frac{1}{3}\begin{bmatrix} 3 & 0 & 0 \\ 0 & 1+1+1 & 1+\omega^2+\omega \\ 0 & 1+\omega+\omega^2 & 1+1+1 \end{bmatrix}$$

$$= \frac{1}{3}\begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

1. $\bar{\omega} = \omega^2$
2. $\bar{\omega^2} = \omega$
3. $\omega^3 = 1$
4. $1+\omega+\omega^2 = 0$

$$AA^\theta = \frac{3}{3}\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$AA^\theta = I$$

Practice Q.9.4 Show that $A = \begin{bmatrix} i & 0 & 0 \\ 0 & 0 & i \\ 0 & i & 0 \end{bmatrix}$ is a skew-Hermitian and unitary.

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Thank You

Engineering Mathematics-1

Lec-10

Unit-1

Matrices

Today's Target

Eigen Values and Eigen Vectors

- ✓ AKTU Questions
- ✓ Practice Questions

Q.10.1 Find the eigen values of the matrix $A = \begin{bmatrix} 1 & 0 & -1 \\ 1 & 2 & 1 \\ 2 & 2 & 3 \end{bmatrix}$

$|A - \lambda I| = 0$

$$\begin{vmatrix} 1-\lambda & 0 & -1 \\ 1 & 2-\lambda & 1 \\ 2 & 2 & 3-\lambda \end{vmatrix} = 0$$

$\lambda^3 - S_1\lambda^2 + S_2\lambda - |A| = 0$

S_1 = sum of diagonal elements

S_2 = sum of Minor of diagonal elements

$S_1 = 1+2+3 = 6$

$S_2 = \begin{vmatrix} 2 & 1 \\ 2 & 3 \end{vmatrix} + \begin{vmatrix} 1 & -1 \\ 2 & 3 \end{vmatrix} + \begin{vmatrix} 1 & 0 \\ 1 & 2 \end{vmatrix}$

$S_2 = (6-2) + (3+2) + (2-0)$

$S_2 = 4 + 5 + 2$

$S_2 = 11$

$|A| = \begin{vmatrix} 1 & 0 & -1 \\ 1 & 2 & 1 \\ 2 & 2 & 3 \end{vmatrix}$

$= 1(6-2) - 0 - 1(2-4)$

$= 4 - 1(-2)$

$= 4 + 2$

$= 6$

$\lambda^3 - 6\lambda^2 + 11\lambda - 6 = 0$

$\lambda = 3, 1, 2$

$\lambda = 1, 2, 3$

Some important properties of eigen values

1. The sum of eigen values of a matrix is equal to trace of the matrix.

$$A = \begin{bmatrix} 5 & 1 & 5 \\ 2 & 1 & 6 \\ 3 & 0 & 7 \end{bmatrix}$$

2. If $\lambda_1, \lambda_2, \lambda_3 \ldots\ldots\ldots \lambda_n$ are the eigen values of A , then the eigen values of

(i) kA are $k\lambda_1, k\lambda_2, k\lambda_3 \ldots\ldots k\lambda_n$

(ii) A^m are $\lambda_1^m, \lambda_2^m, \lambda_3^m \ldots\ldots \lambda_n^m$

(iii) A^{-1} are $\frac{1}{\lambda_1}, \frac{1}{\lambda_2}, \frac{1}{\lambda_3} \ldots\ldots \frac{1}{\lambda_n}$

3. Any square matrix A and its transpose A' have the same eigen values.

4. The eigen values of a diagonal matrix are just the diagonal elements.

$$A = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 5 \end{bmatrix}$$

$$\lambda = 2, 3, 5$$

$$I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\lambda = 1, 1, 1$$

5. The eigen values of a triangular matrix are just the diagonal elements of the matrix.

$$A = \begin{bmatrix} 2 & 3 & 1 \\ 0 & 4 & 6 \\ 0 & 0 & 5 \end{bmatrix}$$

$$\lambda = 2, 4, 5$$

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Q.10.2 Find the eigen values of the matrix $A = \begin{bmatrix} 5 & 4 \\ 1 & 2 \end{bmatrix}$

$$A = \begin{bmatrix} 5 & 4 \\ 1 & 2 \end{bmatrix}$$

$$|A - \lambda I| = 0$$

$$\begin{vmatrix} 5-\lambda & 4 \\ 1 & 2-\lambda \end{vmatrix} = 0$$

$$(5-\lambda)(2-\lambda) - 4 = 0$$

$$10 - 5\lambda - 2\lambda + \lambda^2 - 4 = 0$$

$$\lambda^2 - 7\lambda + 6 = 0$$

$$\lambda = 6, 1$$

AKTU

Q.10.3 Find the eigen values of A^3 where $A = \begin{bmatrix} 3 & 2 \\ 1 & 2 \end{bmatrix}$

$$A = \begin{bmatrix} 3 & 2 \\ 1 & 2 \end{bmatrix}$$

$$|A - \lambda I| = 0$$

$$\begin{vmatrix} 3-\lambda & 2 \\ 1 & 2-\lambda \end{vmatrix} = 0$$

$$(3-\lambda)(2-\lambda) - 2 = 0$$

$$6 - 3\lambda - 2\lambda + \lambda^2 = 0$$

$$\lambda^2 - 5\lambda + 6 = 0$$

$$\lambda = 1, 4$$

Eigen values of A^3 are 1^3 and 4^3

i.e 1 and 64

AKTU

Q.10.4 The matrix A is defined as $A = \begin{bmatrix} 1 & 2 & -3 \\ 0 & 3 & 2 \\ 0 & 0 & -2 \end{bmatrix}$

Find the eigen values of $3A^3 + 5A^2 - 6A + 2I$

$$A = \begin{bmatrix} 1 & 2 & -3 \\ 0 & 3 & 2 \\ 0 & 0 & -2 \end{bmatrix}$$

$\lambda = 1, 3, -2$

$3A^3 + 5A^2 - 6A + 2I$

First eigen value

$= 3(1)^3 + 5(1)^2 - 6\times1 + 2\times1$

$= 3 + 5 - 6 + 2$

$= 4$

Second eigen value

$= 3(3)^3 + 5(3)^2 - 6\times3 + 2\times1$

$= 81 + 45 - 18 + 2$

$= 110$

Third Eigen value

$= 3(-2)^3 + 5(-2)^2 - 6\times(-2) + 2\times1$

$= -24 + 20 + 12 + 2$

$= 10$

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Q.10.5 Two eigen values of the matrix $A = \begin{bmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{bmatrix}$ are equal to 1 each. Find the eigen values of A^{-1}

Let λ be the third eigen value

$$\lambda + 1 + 1 = 2 + 3 + 2$$

$$\lambda + 2 = 7$$

$$\lambda = 5$$

Eigen values of $A = 1, 1, 5$

Eigen values of A^{-1}

$$= 1, 1, \frac{1}{5}$$

Q.10.6 Find the eigen values and eigen vectors of the matrix $A = \begin{bmatrix} 3 & 1 & 4 \\ 0 & 2 & 6 \\ 0 & 0 & 5 \end{bmatrix}$

$|A - \lambda I| = 0$

$$\begin{vmatrix} 3-\lambda & 1 & 4 \\ 0 & 2-\lambda & 6 \\ 0 & 0 & 5-\lambda \end{vmatrix} = 0$$

Expand along C_1

$(3-\lambda)(2-\lambda)(5-\lambda) = 0$

✓ $\lambda = 3$

✓ $\lambda = 2$

✓ $\lambda = 5$

Non-Zero Eigen vectors corresponding to eigen values

$$(A - \lambda I)X = 0$$

$$\begin{bmatrix} 3-\lambda & 1 & 4 \\ 0 & 2-\lambda & 6 \\ 0 & 0 & 5-\lambda \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

When $\lambda = 3$

$$\begin{bmatrix} 0 & 1 & 4 \\ 0 & -1 & 6 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$0 + x_2 + 4x_3 = 0$ — ①

$0 - x_2 + 6x_3 = 0$ — ②

$0 + 0 + 2x_3 = 0$ — ③

$2x_3 = 0$

$\boxed{x_3 = 0}$ Let $x_1 = K_1$

$\boxed{x_2 = 0}$

$$X_1 = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} K_1 \\ 0 \\ 0 \end{bmatrix}$$

$$\boxed{X_1 = K_1 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}}$$

When $\lambda = 2$

$$\begin{bmatrix} 1 & 1 & 4 \\ 0 & 0 & 6 \\ 0 & 0 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$x_1 + x_2 + 4x_3 = 0 \quad \text{— (1)}$$

$$6x_3 = 0$$

$$\boxed{x_3 = 0}$$

$$3x_3 = 0$$

$$x_3 = 0$$

$$x_1 + x_2 + 0 = 0$$

$$x_1 + x_2 = 0$$

$$x_1 = -x_2$$

$$\frac{x_1}{1} = \frac{x_2}{-1} = k_2$$

$$x_1 = k_2$$

$$x_2 = -k_2$$

$$X_2 = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} k_2 \\ -k_2 \\ 0 \end{bmatrix}$$

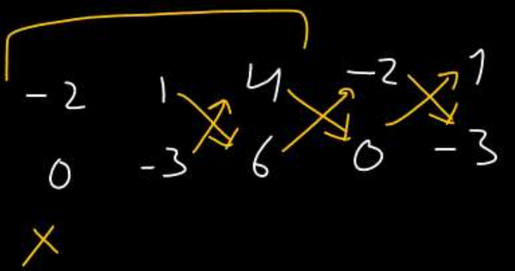
$$\boxed{X_2 = k_2 \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}}$$

When $\lambda = 5$

$$\begin{bmatrix} -2 & 1 & 4 \\ 0 & -3 & 6 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$-2x_1 + x_2 + 4x_3 = 0$$

$$0x_1 - 3x_2 + 6x_3 = 0$$



Apply cross multiplication Method

$$\frac{x_1}{6+12} = \frac{x_2}{0+12} = \frac{x_3}{6-0}$$

$$\frac{x_1}{18} = \frac{x_2}{12} = \frac{x_3}{6}$$

$$\frac{x_1}{3} = \frac{x_2}{2} = \frac{x_3}{1}$$

$$\frac{x_1}{3} = \frac{x_2}{2} = \frac{x_3}{1} = k_3$$

$$x_1 = 3k_3$$

$$x_2 = 2k_3$$

$$x_3 = k_3$$

$$X_3 = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$X_3 = \begin{bmatrix} 3k_3 \\ 2k_3 \\ k_3 \end{bmatrix}$$

$$X_3 = k_3 \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix}$$

Q.10.7 Find all the eigen values and eigen vectors of the matrix $A = \begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix}$

$$|A - \lambda I| = 0$$

$$\begin{vmatrix} -2-\lambda & 2 & -3 \\ 2 & 1-\lambda & -6 \\ -1 & -2 & -\lambda \end{vmatrix} = 0$$

$$\boxed{\lambda^3 - S_1\lambda^2 + S_2\lambda - |A| = 0}$$

$S_1 = -2+1+0 = -1$

$$S_2 = \begin{vmatrix} 1 & -6 \\ -2 & 0 \end{vmatrix} + \begin{vmatrix} -2 & -3 \\ -1 & 0 \end{vmatrix} + \begin{vmatrix} -2 & 2 \\ 2 & 1 \end{vmatrix}$$

$= (0-12) + (0-3) + (-2-4)$

$= -12 - 3 - 6$

$S_2 = -21$

$$|A| = \begin{vmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{vmatrix}$$

$|A| = -2(0-12) - 2(0-6) - 3(-4+1)$

$|A| = 24 + 12 + 9$

$|A| = 45$

$\lambda^3 + \lambda^2 - 21\lambda - 45 = 0$

$\boxed{\lambda = 5, -3, -3}$

Non-zero Eigen vector corresponding to Eigen values

$$(A - \lambda I)X = 0$$

$$\begin{bmatrix} -2-\lambda & 2 & -3 \\ 2 & 1-\lambda & -6 \\ -1 & -2 & -\lambda \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

When $\lambda = -3$

$$\begin{bmatrix} 1 & 2 & -3 \\ 2 & 4 & -6 \\ -1 & -2 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & -3 \\ 2 & 4 & -6 \\ -1 & -2 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$R_2 \longrightarrow R_2 + -2R_1$

$R_3 \longrightarrow R_3 + R_1$

$$\begin{bmatrix} 1 & 2 & -3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$x_1 + 2x_2 - 3x_3 = 0$ — (1)

Let $x_2 = k_1$

$x_3 = k_2$

$x_1 + 2k_1 - 3k_2 = 0$

$x_1 = 3k_2 - 2k_1$

$$X_1 = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 3k_2 - 2k_1 \\ k_1 \\ k_2 \end{bmatrix}$$

$$X_1 = \begin{bmatrix} 3k_2 - 2k_1 \\ k_1 + 0 \\ k_2 + 0 \end{bmatrix}$$

$$X_1 = \begin{bmatrix} -2k_1 \\ k_1 \\ 0 \end{bmatrix} + \begin{bmatrix} 3k_2 \\ k \\ k_2 \end{bmatrix}$$

$$X_2 = k_1 \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} + k_2 \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix}$$

$$X_4 = k_4 \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}$$

$$X_2 = k_2 \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix}$$

Practice Q. 10.1 The matrix $A = \begin{bmatrix} 1 & 0 \\ 2 & 4 \end{bmatrix}$ is given. Find the eigen values of $4A^{-1} + 3A + 2I$

Ans. 6,15

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Practice Q. 10.2 Find all the eigen values and eigen vectors of the matrix

$$A = \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$$

Ans. 0,3,15

AKTU

Practice Q. 10.3 Find all the eigen values and eigen vectors of the matrix

$$A = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$$

Ans. 2,2,8

Thank You

Engineering Mathematics-1

Lec-11

Unit-1

Matrices

Today's Target

Cayley-Hamilton Theorem

- ✓ AKTU Questions
- ✓ Practice Questions

Cayley-Hamilton Theorem

Every square matrix satisfies its own characteristics equation.

Q.11.1 Verify Cayley-Hamilton theorem for a matrix $A = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$ and hence find the inverse of the matrix.

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The characteristic equation is

$$|A - \lambda I| = 0$$

$$\begin{vmatrix} 2-\lambda & -1 & 1 \\ -1 & 2-\lambda & -1 \\ 1 & -1 & 2-\lambda \end{vmatrix} = 0$$

$$\lambda^3 - S_1\lambda^2 + S_2\lambda - |A| = 0$$

S_1 = Sum of diagonal elements

S_2 = Sum of minor of diagonal elements

$$S_2 = \begin{vmatrix} 2 & -1 \\ -1 & 2 \end{vmatrix} + \begin{vmatrix} 2 & 1 \\ 1 & 2 \end{vmatrix} + \begin{vmatrix} 2 & -1 \\ -1 & 2 \end{vmatrix}$$

$$S_2 = (4-1) + (4-1) + (4-1)$$

$$S_2 = 3 + 3 + 3 = 9$$

$$S_1 = 2 + 2 + 2 = 6$$

$$|A| = \begin{vmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{vmatrix}$$

$$= 2\begin{vmatrix} 2 & -1 \\ -1 & 2 \end{vmatrix} + 1\begin{vmatrix} -1 & -1 \\ 1 & 2 \end{vmatrix} + 1\begin{vmatrix} -1 & 2 \\ 1 & -1 \end{vmatrix}$$

$$= 2(4-1) + 1(-2+1) + 1(1-2)$$

$$= 2 \times 3 - 1 - 1$$

$$= 6 - 1 - 1$$

$$= 4$$

$$\lambda^3 - 6\lambda^2 + 9\lambda - 4 = 0$$

By cayley Hamilton theorem

$$A^3 - 6A^2 + 9A - 4I = 0$$

$$A^3 - 6A^2 + 9A - 4I = 0$$

$$A^3 = A \times A^2$$

$$A^2 = A \times A$$

$$A^2 = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix} \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} 4+1+1 & -2-2-1 & 2+1+2 \\ -2-2-1 & 1+4+1 & -1-2-2 \\ 2+1+2 & -1-2-2 & 1+1+4 \end{bmatrix}$$

$$= \begin{bmatrix} 6 & -5 & 5 \\ -5 & 6 & -5 \\ 5 & -5 & 6 \end{bmatrix}$$

$$A^3 = A \times A^2$$

$$A^3 = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix} \begin{bmatrix} 6 & -5 & 5 \\ -5 & 6 & -5 \\ 5 & -5 & 6 \end{bmatrix}$$

$$= \begin{bmatrix} 12+5+5 & -10-6-5 & 10+5+6 \\ -6-10-5 & 5+12+5 & -5-10-6 \\ 6+5+10 & -5-6-10 & 5+5+12 \end{bmatrix}$$

$$= \begin{bmatrix} 22 & -21 & 21 \\ -21 & 22 & -21 \\ 21 & -21 & 22 \end{bmatrix}$$

$$A^3 - 6A^2 + 9A - 4I =$$

$$\begin{bmatrix} 22 & -21 & 21 \\ -21 & 22 & -21 \\ 21 & -21 & 22 \end{bmatrix} - 6\begin{bmatrix} 6 & -5 & 5 \\ -5 & 6 & -5 \\ 5 & -5 & 6 \end{bmatrix} + 9\begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix} - 4\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 22 & -21 & 21 \\ -21 & 22 & -21 \\ 21 & -21 & 22 \end{bmatrix} + \begin{bmatrix} -36 & 30 & -30 \\ 30 & -36 & 30 \\ -30 & 30 & -36 \end{bmatrix} + \begin{bmatrix} 18 & -9 & 9 \\ -9 & 18 & -9 \\ 9 & -9 & 18 \end{bmatrix} + \begin{bmatrix} -4 & 0 & 0 \\ 0 & -4 & 0 \\ 0 & 0 & -4 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$= O$$

Hence, cayley Hamilton theorem is verified

$$A^3 - 6A^2 + 9A - 4I = O$$

Pre-Multiply by A^{-1}

$$A^{-1}A^3 - 6A^{-1}A^2 + 9A^{-1}A - 4A^{-1}I = A^{-1} \times O$$

$$A^2 - 6A + 9I - 4A^{-1} = O$$

$$4A^{-1} = A^2 - 6A + 9I$$

$$= \begin{bmatrix} 6 & -5 & 5 \\ -5 & 6 & -5 \\ 5 & -5 & 6 \end{bmatrix} - 6\begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix} + 9\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 6 & -5 & 5 \\ -5 & 6 & -5 \\ 5 & -5 & 6 \end{bmatrix} + \begin{bmatrix} -12 & 6 & -6 \\ 6 & -12 & 6 \\ -6 & 6 & -12 \end{bmatrix} + \begin{bmatrix} 9 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 9 \end{bmatrix}$$

$$= \begin{bmatrix} 3 & 1 & -1 \\ 1 & 3 & 1 \\ -1 & 1 & 3 \end{bmatrix}$$

$$A^{-1} = \frac{1}{4}\begin{bmatrix} 3 & 1 & -1 \\ 1 & 3 & 1 \\ -1 & 1 & 3 \end{bmatrix}$$

Practice Q.11.1 Verify Cayley-Hamilton theorem for a matrix and hence find the inverse of the matrix.

$$A = \begin{bmatrix} 4 & 3 & 1 \\ 2 & 1 & -2 \\ 1 & 2 & 1 \end{bmatrix}$$

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Q.11.2 Verify Cayley-Hamilton theorem for the matrix. $A = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix}$

And hence compute A^{-1}. Also evaluate $A^8 - 5A^7 + 7A^6 - 3A^5 + A^4 - 5A^3 + 8A^2 - 2A + I$.

$|A - \lambda I| = 0$

$$\begin{vmatrix} 2-\lambda & 1 & 1 \\ 0 & 1-\lambda & 0 \\ 1 & 1 & 2-\lambda \end{vmatrix} = 0$$

$\lambda^3 - S_1\lambda^2 + S_2\lambda - |A| = 0$

$S_1 = 2+1+2 = 5$

$S_2 = \begin{vmatrix} 1 & 0 \\ 1 & 2 \end{vmatrix} + \begin{vmatrix} 2 & 1 \\ 1 & 2 \end{vmatrix} + \begin{vmatrix} 2 & 1 \\ 0 & 1 \end{vmatrix}$

$= (2-0) + (4-1) + (2-0)$

$= 2 + 3 + 2$

$= 7$

$|A| = \begin{vmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{vmatrix}$

$= 2\begin{vmatrix} 1 & 0 \\ 1 & 2 \end{vmatrix} - 0 + 1\begin{vmatrix} 1 & 1 \\ 1 & 0 \end{vmatrix}$

$= 2(2-0) + 1(0-1)$

$= 4 - 1$

$= 3$

$\lambda^3 - 5\lambda^2 + 7\lambda - 3 = 0$

By cayley Hamilton theorem

$A^3 - 5A^2 + 7A - 3I = 0$

$A^2 = A \times A$

$= \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix} \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix}$

$= \begin{bmatrix} 4+0+1 & 2+1+1 & 2+0+2 \\ 0+0+0 & 0+1+0 & 0+0+0 \\ 2+0+2 & 1+1+2 & 1+0+4 \end{bmatrix}$

$A^2 = \begin{bmatrix} 5 & 4 & 4 \\ 0 & 1 & 0 \\ 4 & 4 & 5 \end{bmatrix}$

$$A^3 = A \times A^2$$

$$= \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix} \begin{bmatrix} 5 & 4 & 4 \\ 0 & 1 & 0 \\ 4 & 4 & 5 \end{bmatrix}$$

$$= \begin{bmatrix} 10+0+4 & 8+1+4 & 8+0+5 \\ 0+0+0 & 0+1+0 & 0+0+0 \\ 5+0+8 & 4+1+8 & 4+0+10 \end{bmatrix}$$

$$A^3 = \begin{bmatrix} 14 & 13 & 13 \\ 0 & 1 & 0 \\ 13 & 13 & 14 \end{bmatrix}$$

$$A^3 - 5A^2 + 7A - 3I$$

$$\begin{bmatrix} 14 & 13 & 13 \\ 0 & 1 & 0 \\ 13 & 13 & 14 \end{bmatrix} - 5\begin{bmatrix} 5 & 4 & 4 \\ 0 & 1 & 0 \\ 4 & 4 & 5 \end{bmatrix} + 7\begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix} - 3\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 14 & 13 & 13 \\ 0 & 1 & 0 \\ 13 & 13 & 14 \end{bmatrix} + \begin{bmatrix} -25 & -20 & -20 \\ 0 & -5 & 0 \\ -20 & -20 & -25 \end{bmatrix} + \begin{bmatrix} 14 & 7 & 7 \\ 0 & 7 & 0 \\ 7 & 7 & 14 \end{bmatrix} + \begin{bmatrix} -3 & 0 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & -3 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = O$$

Hence cayley Hamilton theorem is verified

Now,

$$A^3 - 5A^2 + 7A - 3I = O$$

Pre-Multiply by A^{-1}

$$A^{-1}A^3 - 5A^{-1}A^2 + 7A^{-1}A - 3A^{-1}I = O \times A^{-1}$$

$$A^2 - 5A + 7I - 3A^{-1} = O$$

$$3A^{-1} = A^2 - 5A + 7I$$

$$= \begin{bmatrix} 5 & 4 & 4 \\ 0 & 1 & 0 \\ 4 & 4 & 5 \end{bmatrix} - 5\begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix} + 7\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$3A^{-1} = \begin{bmatrix} 5 & 4 & 4 \\ 0 & 1 & 0 \\ 4 & 4 & 5 \end{bmatrix} + \begin{bmatrix} -10 & -5 & -5 \\ 0 & -5 & 0 \\ -5 & -5 & -10 \end{bmatrix} + \begin{bmatrix} 7 & 0 & 0 \\ 0 & 7 & 0 \\ 0 & 0 & 7 \end{bmatrix}$$

$$3A^{-1} = \begin{bmatrix} 2 & -1 & -1 \\ 0 & 3 & 0 \\ -1 & -1 & 2 \end{bmatrix}$$

$$A^{-1} = \frac{1}{3}\begin{bmatrix} 2 & -1 & -1 \\ 0 & 3 & 0 \\ -1 & -1 & 2 \end{bmatrix}$$

$$A^3 - 5A^2 + 7A - 3I = 0$$

$7A^2 + A^2$, $-3A + A$

$$= A^8 - 5A^7 + 7A^6 - 3A^5 + A^4 - 5A^3 + 8A^2 - 2A + I$$

$$= A^5(A^3 - 5A^2 + 7A - 3I) + A(A^3 - 5A^2 + 7A - 3I) + A^2 + A + I$$

$$= A^5 \times 0 + A \times 0 + A^2 + A + I$$

$$= A^2 + A + I$$

$$= \begin{bmatrix} 5 & 4 & 4 \\ 0 & 1 & 0 \\ 4 & 4 & 5 \end{bmatrix} + \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix} + \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 8 & 5 & 5 \\ 0 & 3 & 0 \\ 5 & 5 & 8 \end{bmatrix}$$

Practice Q.11.2 Verify Cayley-Hamilton theorem for the matrix. $A = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$

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Hence compute A^{-1}. Also evaluate $A^6 - 6A^5 + 9A^4 - 2A^3 - 12A^2 + 23A - 9I$

Practice Q.11.3 If $A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$ find A^{-1} and A^4 using Cayley-Hamilton theorem.

Practice Q.11.4 Verify Cayley-Hamilton theorem for the matrix. $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 6 \end{bmatrix}$

Thank You